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CASE INCLUDING STRUCTURE FOR RECEIVING ELECTRONIC PEN

Abstract

A case detachably coupled to an electronic device is provided. The case includes a plate supporting the electronic device and a storing member for storing at least a portion of an electronic pen disposed on a surface of the plate, wherein the storing member includes a receiving space receiving an end of the electronic pen, a first actuating portion applying a force in a direction toward the receiving space to another end of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space, and a second actuating portion, disposed between the receiving space and the first actuating portion, and applying a force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member, and wherein a portion of the storing member is open toward an outside of the storing member such that at least a portion of the electronic pen is exposed outside the storing member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365 (c), of an International application No. PCT/KR2023/017114, filed on Oct. 31, 2023, which is based on and claims the benefit of a Korean patent application number 10-2023-0004889, filed on Jan. 12, 2023, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2023-0015506, filed on Feb. 6, 2023, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to a case including a structure for receiving an electronic pen.

2. Description of Related Art

[0003] An electronic device may be exposed to various environments, by being carried by a user. A case may protect the electronic device exposed to various environments, by being detachably coupled to the electronic device. For example, the case may protect the electronic device from an impact transmitted to the electronic device from the outside of the electronic device. In order to meet a demand of the user, the case may have various functions in addition to a function for protecting the electronic device.

[0004] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

[0005] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide a case for including structure for receiving electronic pen.

[0006] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

[0007] In accordance with an aspect of the disclosure, a case detachably coupled to an electronic device is provided. The case includes a plate supporting the electronic device, a storing member for storing at least a portion of an electronic pen disposed on a surface of the plate, wherein the storing member includes a receiving space receiving an end of the electronic pen, a first actuating portion applying a force in a direction toward the receiving space to another end of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space, and a second actuating portion, disposed between the receiving space and the first actuating portion, and applying a force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member, and wherein a portion of the storing member opens toward an outside of the storing member such that at least a portion of the electronic pen is exposed outside the storing member.

[0008] In accordance with another aspect of the disclosure, a case detachably coupled to an

electronic device is provided. The case includes a plate supporting the electronic device, and a storing member for storing at least a portion of an electronic pen disposed on a surface of the plate, wherein the storing member includes a seating portion, having a first thickness, on which the electronic pen is seated a receiving space receiving an end of the electronic pen, a first protruding portion, having a second thickness greater than the first thickness, surrounding the receiving space, and connected to an end of the seating portion, a first actuating portion applying a force in a direction toward the receiving space to another end of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space, a second protruding portion, having a third thickness greater than the first thickness, surrounding the first actuating portion, and connected to another end of the seating portion opposite to the end of the seating portion, and a second actuating portion, disposed in the seating portion, applying a force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member.

[0009] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0011] FIG. 1A is a perspective view illustrating an example in which a case and an electronic device are coupled according to an embodiment of the disclosure;

[0012] FIG. 1B is a top plan view illustrating an example in which a case and an electronic device are coupled according to an embodiment of the disclosure;

[0013] FIG. 1C is an exploded perspective view of a case according to an embodiment of the disclosure;

[0014] FIG. 2 is a cross-sectional view illustrating an example of a case cut along line A-A' of FIG. 1B according to an embodiment of the disclosure;

[0015] FIG. 3A is a perspective view of a case according to an embodiment of the disclosure;

[0016] FIG. 3B is a cross-sectional view illustrating an example of a case cut along line B-B' of FIG. 1B according to an embodiment of the disclosure;

[0017] FIG. 4A is a top plan view of a case according to an embodiment of the disclosure;

[0018] FIG. 4B is a cross-sectional view illustrating an example of a case cut along line C-C' of FIG. 4A according to an embodiment of the disclosure; and

[0019] FIG. 5 is a cross-sectional view illustrating an example of a case according to an embodiment of the disclosure.

[0020] The same reference numerals are used to represent the same elements throughout the drawings.

DETAILED DESCRIPTION

[0021] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

[0022] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

[0023] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0024] It should be appreciated that the blocks in each flowchart and combinations of the flowcharts may be performed by one or more computer programs which include instructions. The entirety of the one or more computer programs may be stored in a single memory device or the one or more computer programs may be divided with different portions stored in different multiple memory devices.

[0025] Any of the functions or operations described herein can be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry performing processing and includes circuitry like an application processor (AP, e.g. a central processing unit (CPU)), a communication processor (CP, e.g., a modem), a graphics processing unit (GPU), a neural processing unit (NPU) (e.g., an artificial intelligence (AI) chip), a wireless fidelity (Wi-Fi) chip, a Bluetooth® chip, a global positioning system (GPS) chip, a near field communication (NFC) chip, connectivity chips, a sensor controller, a touch controller, a fingerprint sensor controller, a display driver integrated circuit (IC), an audio CODEC chip, a universal serial bus (USB) controller, a camera controller, an image processing IC, a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.

[0026] FIG. 1A is a perspective view illustrating an example in which a case and an electronic device are coupled according to an embodiment of the disclosure.

[0027] FIG. 1B is a top plan view illustrating an example in which a case and an electronic device are coupled according to an embodiment of the disclosure.

[0028] FIG. 1C is an exploded perspective view of a case according to an embodiment of the disclosure.

[0029] Referring to FIGS. 1A, 1B, and 1C, according to an embodiment, an electronic device **100** may be detachably coupled to a case **200**. The electronic device **100** may be disposed on the case **200**. According to another embodiment, the electronic device **100** may include a housing **110** and a display **120**.

[0030] According to yet another embodiment, the housing **110** may define at least a portion of an outer surface of the electronic device **100**. For example, the housing **110** may define at least a portion of a rear surface and a side surface of the electronic device **100**. The rear surface of the electronic device **100** may mean a surface opposite to a front surface of the electronic device **100** facing a user while the user is provided with visual information through the electronic device **100**. The side surface of the electronic device **100** may mean a surface connecting the front surface of the electronic device **100** and the rear surface of the electronic device **100**. The housing **110** may provide an internal space in which at least a portion of components of the electronic device **100** are disposed. For example, the housing **110** may receive various electronic components of the electronic device **100** to provide various functions to the user. According to yet another embodiment, the housing **110** may directly contact the case **200** in a state in which the electronic device **100** and the case **200** are coupled to each other. For example, the housing **110** may be disposed on the case **200** in the state in which the electronic device **100** and the case **200** are coupled to each other.

[0031] According to yet another embodiment, the display **120** may provide visual information to the user. The display **120** may be configured to emit light for providing visual content to the user.

According to yet another embodiment, the display **120** may be disposed on the housing **110**. The display **120** may be coupled to the housing **110**. For example, the display **120** may define at least a portion of the front surface of the electronic device **100** by being disposed on the housing **110**. For example, the display **120** may face the user while the user is provided with visual information through the electronic device **100**.

[0032] According to yet another embodiment, an electronic pen **130** may be configured to transmit a signal corresponding to an input of the user to the electronic device **100**. The electronic pen **130** may be configured to transmit a signal indicating an input of the user received on the display **120** to the electronic device **100**. The electronic pen **130** may be configured to transmit a signal indicating an input of the user received above the display **120** to the electronic device **100**. For example, the electronic pen **130** may be configured to transmit an electromagnetic signal corresponding to an input of the user to the electronic device **100**. For example, the electronic pen **130** may include a coil (not illustrated) for transmitting a user input to the electronic device **100** through an electromagnetic signal. For example, the electronic device **100** may include an electromagnetic induction panel (not illustrated) for receiving a user input from the electronic pen **130** through an electromagnetic signal. For example, the electronic pen **130** may include an electro-magnetic resonance (EMR) method. For example, the electromagnetic induction panel of the electronic device **100** may transmit a first electromagnetic signal that causes electromagnetic resonance in the electronic pen **130** to the electronic pen **130**. The electronic pen **130** may obtain a second electromagnetic signal by a current caused by electromagnetic induction in the electronic pen **130** based on receiving the first electromagnetic signal. The electronic pen **130** may transmit the second electromagnetic signal causing electromagnetic resonance in the electronic device **100** to the electromagnetic induction panel of the electronic device **100**. The electronic device **100** may obtain a third electromagnetic signal by a current caused by electromagnetic induction based on receiving the second electromagnetic signal through the electromagnetic induction panel. The electronic device **100** may identify a type of the user input from the electronic pen **130** based on identifying a phase of the third electromagnetic signal. For example, the electronic device **100** may identify the user input from the electronic pen **130** as a touch input on the display **120** based on identifying that the phase of the third electromagnetic signal is less than or equal to a first threshold value. For example, the electronic device **100** may identify the user input from the electronic pen **130** as a hovering input above the display **120** based on identifying that the phase of the third electromagnetic signal is greater than the first threshold value. The electronic device **100** may obtain information on a position of the electronic pen **130** based on the third electromagnetic signal. For example, the information on the position of the electronic pen **130** may include coordinates on the display **120** corresponding to the position of the electronic pen **130** above the display **120**. However, an operation method of the electronic pen **130** is not limited thereto. For example, the electronic device **100** may identify the user input from the electronic pen **130** through a capacitive sensor in the display **120**. For example, the electronic pen **130** may include an active electrostatic (AES) type, a pressure-sensitive type, or an infrared/ultrasonic type pen.

[0033] According to yet another embodiment, the electronic pen **130** may substantially have a shape of a pen. For example, the electronic pen **130** may be referred to as a stylus pen. For example, an end **130a** of the electronic pen **130** may have a shape of a pen tip. For example, the electronic pen **130**, and the end **130a** of the electronic pen **130** may include a material having flexibility. For example, another end **130b** of the electronic pen **130** may be opposite to the end **130a** of the electronic pen **130**. For example, a length of the electronic pen **130** may be defined as a distance between the end **130a** of the electronic pen **130** and the other end **130b** of the electronic pen **130**. A length of a component may mean a distance in a direction (e.g., a +y direction or a-y direction) parallel to a direction (e.g., the -y direction) from the end **130a** of the electronic pen **130** toward the other end **130b** of the electronic pen **130**, and the corresponding expression may be utilized substantially the same below unless otherwise stated.

[0034] According to yet another embodiment, the case **200** may protect the electronic device **100**, by being detachably coupled to the electronic device **100**. The case **200** may buffer an impact transmitted to the electronic device **100** from the outside of the electronic device **100**. According to yet another embodiment, the case **200** may include a plate **210** and/or a storing member **220**.

[0035] According to yet another embodiment, the plate **210** may define at least a portion of an outer surface of the case **200**. The plate **210** may support the electronic device **100**. The plate **210** may contact the electronic device **100**. For example, a portion of a surface **210a** of the plate **210** may face the electronic device **100**. A portion of the surface **210a** of the plate **210** may be in contact with the housing **110** of the electronic device **100**. According to yet another embodiment, a size of the plate **210** may be larger than a size of the electronic device **100**. The plate **210** may include a first portion **211**, a second portion **212**, and/or a third portion **213**. According to yet another embodiment, the first portion **211** may support the electronic device **100**. The first portion **211** may be in contact with the electronic device **100**. The electronic device **100** may be disposed on the first portion **211**. For example, the electronic device **100** may be disposed on an inner surface **211a** (e.g., the first inner surface **211a**) of the first portion **211**. For example, the inner surface **211a** of the first portion **211** may face the rear surface of the electronic device **100**. The second portion **212** may be movable with respect to the first portion **211**. For example, the second portion **212** may be rotatable with respect to the first portion **211**. For example, the second portion **212** may be foldable with respect to the first portion **211**. An inner surface **212a** (e.g., the second inner surface **212a**) of the second portion **212** may be spaced apart from the inner surface **211a** (e.g., the first inner surface **211a**) of the first portion **211**. The third portion **213** may be disposed between the first portion **211** and the second portion **212**. The third portion **213** may connect the first portion **211** and the second portion **212**. For example, an end of the third portion **213** may be connected to the first portion **211**, and another end of the third portion **213** may be connected to the second portion **212**. An inner surface **213a** (e.g., the third inner surface **213a**) of the third portion **213** may be disposed between the inner surface **211a** (e.g., the first inner surface **211a**) of the first portion **211** and the inner surface **212a** (e.g., the second inner surface **212a**) of the second portion **212**. The inner surface **213a** of the third portion **213** may connect the inner surface **211a** of the first portion **211** and the inner surface **212a** of the second portion **212**. The surface **210a** of the plate **210** may include the inner surface **211a** of the first portion **211**, the inner surface **212a** of the second portion **212**, and the inner surface **213a** of the third portion **213**. The surface **210a** of the plate **210** may be defined by the inner surface **211a** of the first portion **211**, the inner surface **212a** of the second portion **212**, and the inner surface **213a** of the third portion **213**.

[0036] According to yet another embodiment, as the first portion **211** and the second portion **212** are movable with respect to each other, a state of the case **200** may include a first state and a second state. In the first state of the case **200**, a direction (e.g., a +z direction) in which the inner surface **211a** of the first portion **211** faces may be substantially the same as a direction (e.g., the +z direction) in which the inner surface **212a** of the second portion **212** faces. In the first state of the case **200**, the display **120** may be visually exposed to the outside of the case **200**. In the first state of the case **200**, the display **120** may not be covered by the second portion **212**. As the second portion **212** moves with respect to the first portion **211**, the state of the case **200** may be changed from the first state to the second state. In the second state of the case **200**, the inner surface **211a** of the first portion **211** and the inner surface **212a** of the second portion **212** may face each other. For example, in the second state of the case **200**, the direction in which the inner surface **211a** of the first portion **211** faces and the direction in which the inner surface **212a** of the second portion **212** faces may be opposite to each other. In the second state of the case **200**, the display **120** may be covered by the second portion **212**. As the second portion **212** moves with respect to the first portion **211**, the state of the case **200** may be changed from the second state to the first state.

[0037] According to yet another embodiment, the storing member **220** may accommodate at least a portion of the electronic pen **130**. The storing member **220** may support the electronic pen **130**. The

electronic pen **130** may be disposed on the storing member **220**. For example, the electronic pen **130** may be disposed on a surface **220a** of the storing member **220**. According to yet another embodiment, the storing member **220** may be disposed on the plate **210**. For example, the storing member **220** may be disposed on the surface **210a** of the plate **210**. The storing member **220** may be disposed between the first portion **211** and the second portion **212**. The storing member **220** may be disposed on the third portion **213**. For example, the storing member **220** may be disposed on the inner surface **213a** of the third portion **213**. According to yet another embodiment, the storing member **220** may include a seating portion **221**, a first protruding portion **222**, a second protruding portion **223**, a receiving space **224**, a first actuating portion **225**, a second actuating portion **226**, and/or a supporting member **227**.

[0038] According to yet another embodiment, the seating portion **221** may support the electronic pen **130**. The electronic pen **130** may be seated on the seating portion **221**. The seating portion **221** may be disposed between the first protruding portion **222** and the second protruding portion **223**. The seating portion **221** may define a step between the first protruding portion **222** and the second protruding portion **223**. For example, the seating portion **221** may have a thickness thinner than a thickness of each of the first protruding portion **222** and the second protruding portion **223**. A thickness of a component may mean a distance in a direction (e.g., the +z direction or a -z direction) parallel to a direction (e.g., the +z direction) in which the surface **210a** of the plate **210** faces in the first state of the case **200**, and the corresponding expression may be utilized substantially the same below unless otherwise stated. According to yet another embodiment, a length of the seating portion **221** may correspond to a length of the electronic pen **130**. For example, the length of the seating portion **221** may be substantially the same as the length of the electronic pen **130**, but is not limited thereto. For example, the length of the seating portion **221** may be longer or shorter than the length of the electronic pen **130**.

[0039] According to yet another embodiment, the first protruding portion **222** may receive the end **130a** of the electronic pen **130**. For example, the first protruding portion **222** may surround the end **130a** of the electronic pen **130**. For example, the first protruding portion **222** may envelop the end **130a** of the electronic pen **130**. The first protruding portion **222** may form the receiving space **224**. The first protruding portion **222** may surround the receiving space **224**. The first protruding portion **222** may envelop the receiving space **224**. The first protruding portion **222** may have a thickness thicker than a thickness of the seating portion **221**. The first protruding portion **222** may be connected to the seating portion **221**. For example, the first protruding portion **222** may be connected to an end of the seating portion **221**.

[0040] According to yet another embodiment, the second protruding portion **223** may be connected to the seating portion **221**. The second protruding portion **223** may be connected to another end of the seating portion **221** opposite to the end of the seating portion **221** connected to the first protruding portion **222**. According to yet another embodiment, the second protruding portion **223** may receive the first actuating portion **225**. For example, the second protruding portion **223** may surround the first actuating portion **225**. For example, the second protruding portion **223** may envelop the first actuating portion **225**. The second protruding portion **223** may have a thickness thicker than the thickness of the seating portion **221**. For example, the thickness of the second protruding portion **223** may be substantially the same as the thickness of the first protruding portion **222**, but is not limited thereto. For example, the thickness of the second protruding portion **223** may be different from the thickness of the first protruding portion **222**.

[0041] According to yet another embodiment, the receiving space **224** may receive a portion of the electronic pen **130**. The receiving space **224** may receive the end **130a** of the electronic pen **130**. The electronic pen **130** may be inserted into the receiving space **224**. In a state in which the end **130a** of the electronic pen **130** is disposed in the receiving space **224**, a portion of the electronic pen **130** may be surrounded (or enveloped) by the first protruding portion **222**. For example, the receiving space **224** may be defined as an empty space disposed inside the first protruding portion

222.

[0042] According to yet another embodiment, the first actuating portion **225** may detachably couple the electronic pen **130** and the case **200**. The first actuating portion **225** may insert a portion of the electronic pen **130** into the receiving space **224**. The first actuating portion **225** may apply a force in a first direction (e.g., the +y direction) toward the receiving space **224** to the other end **130b** of the electronic pen **130**. The first actuating portion **225** may move the electronic pen **130** in the first direction (e.g., the +y direction). The first actuating portion **225** may reduce movement in a second direction (e.g., the -y direction) opposite to the first direction (e.g., the +y direction) of the electronic pen **130**. For example, the first actuating portion **225** may apply the force in the first direction (e.g., the +y direction) from the end **130a** of the electronic pen **130** toward the other end **130b** of the electronic pen **130** to the other end **130b** of the electronic pen **130**. For example, the first actuating portion **225** may be described as a repelling portion (or an repelling structure) that applies a force to push the other end **130b** of the electronic pen **130** in a direction (e.g., the +y direction) away from the first actuating portion **225**. For example, the first actuating portion **225** may be referred to as a first coupling portion that applies a force in the direction (e.g., the +y direction) away from the first actuating portion **225** to the other end **130b** of the electronic pen **130**. According to yet another embodiment, the first actuating portion **225** may be disposed in the storing member **220**. The first actuating portion **225** may be disposed in the second protruding portion **223**. The first actuating portion **225** may be spaced apart from the surface **210a** of the plate **210** in the second protruding portion **223**. For example, the first actuating portion **225** may be spaced apart from the inner surface **213a** of the third portion **213** in the second protruding portion **223**.

[0043] According to yet another embodiment, the second actuating portion **226** may seat the electronic pen **130** on the storing member **220**. The second actuating portion **226** may apply a force to the electronic pen **130** such that the electronic pen **130** maintains a state in which the electronic pen **130** is seated on the storing member **220**. For example, the second actuating portion **226** may apply a force in a third direction (e.g., the -z direction) toward the surface **210a** of the plate **210** to the electronic pen **130**. For example, the second actuating portion **226** may apply the force in the third direction (e.g., the -z direction) approaching the surface **210a** of the plate **210** to the electronic pen **130**. For example, the second actuating portion **226** may apply the force in the third direction (e.g., the -z direction) opposite to a direction (e.g., the +z direction) in which the surface **210a** of the plate **210** faces in the first state of the case **200** to the electronic pen **130**. According to yet another embodiment, the second actuating portion **226** may be disposed in the storing member **220**. The second actuating portion **226** may be surrounded by the seating portion **221** of the storing member **220**. The second actuating portion **226** may be enveloped by the seating portion **221** of the storing member **220**. The second actuating portion **226** may be attached on the surface **210a** of the plate **210**. For example, the second actuating portion **226** may be attached on the inner surface **213a** of the third portion **213** in the seating portion **221**. For example, the second actuating portion **226** may be received inside a groove formed by recessing the surface **210a** of the plate **210** inwardly. For example, the second actuating portion **226** may be referred to as a second coupling portion that applies a force in a direction (e.g., the -z direction) toward the surface **210a** of the plate to the electronic pen **130**.

[0044] According to yet another embodiment, the supporting member **227** may space apart the first actuating portion **225** from the surface **210a** of the plate **210**. For example, the first actuating portion **225** may be spaced apart from the inner surface **213a** of the third portion **213** by the supporting member **227**. The supporting member **227** may be disposed between the surface **210a** of the plate **210** and the first actuating portion **225**. For example, the supporting member **227** may be in contact with the surface **210a** of the plate **210** and the first actuating portion **225**.

[0045] According to yet another embodiment, the electronic pen **130** may be easily coupled to the case **200** by the first actuating portion **225** and the second actuating portion **226**. For example, a

process in which the electronic pen **130** is coupled to the case **200** by the first actuating portion **225** and the second actuating portion **226** may be described with reference to FIG. 2.

[0046] FIG. 2 is a cross-sectional view illustrating an example of a case cut along line A-A' of FIG. 1B according to an embodiment of the disclosure.

[0047] Referring to FIG. 2, according to an embodiment, a first actuating portion **225** may include a first magnet **225a**. The first magnet **225a** may be disposed in a second protruding portion **223**.

[0048] According to another embodiment, an electronic pen **130** may include a second magnet **131** corresponding to the first magnet **225a**. The second magnet **131** may be disposed in the electronic pen **130**. The second magnet **131** may be closer to another end **130b** from among an end **130a** and the other end **130b** of the electronic pen **130**. For example, the second magnet **131** may be disposed in the other end **130b** of the electronic pen **130**.

[0049] According to yet another embodiment, the first actuating portion **225** may insert the end **130a** of the electronic pen **130** into a receiving space **224**. The first magnet **225a** of the first actuating portion **225** may insert the end **130a** of the electronic pen **130** into the receiving space **224** by interacting with the second magnet **131**. For example, the first magnet **225a** may apply a force in a first direction (e.g., a +y direction) to the other end **130b** of the electronic pen **130** by exerting a repulsive force to the second magnet **131**. The electronic pen **130** may move along the first direction (e.g., the +y direction) by receiving the force in the first direction (e.g., the +y direction) from the first magnet **225a**. As the electronic pen **130** moves in the first direction (e.g., the +y direction), the end **130a** of the electronic pen **130** may be inserted into the receiving space **224**. The first actuating portion **225** may provide convenience for a user to easily couple the electronic pen **130** to a case **200** by providing the force in the first direction to the electronic pen **130**.

[0050] According to yet another embodiment, the first actuating portion **225** may maintain a state in which the electronic pen **130** is inserted into the receiving space **224**. For example, in case that the electronic pen **130** moves in a second direction (e.g., a -y direction) opposite the first direction (e.g., the +y direction), the electronic pen **130** may be pulled out from the receiving space **224**. When the electronic pen **130** is pulled out from the receiving space **224**, the electronic pen **130** may be separated from the case **200** regardless of an intention of the user. When the electronic pen **130** is separated from the case **200** regardless of the intention of the user, the user may lose the electronic pen **130**. The case **200** according to yet another embodiment may maintain a position of the electronic pen **130** in the case **200** by the first actuating portion **225** that applies the force in the first direction toward the receiving space **224** to the electronic pen **130**. As the position of the electronic pen **130** in the case **200** is maintained, a possibility of the electronic pen **130** being lost may be reduced.

[0051] According to yet another embodiment, a first distance d1 between a surface **220a** of a storing member **220** in contact with the electronic pen **130** and the first magnet **225a** of the first actuating portion **225** may be different from a second distance d2 between the surface **220a** of the storing member **220** and the second magnet **131** in the electronic pen **130**. The surface **220a** of the storing member **220** may face a fourth direction (e.g., a +z direction) opposite to a third direction (e.g., a -z direction) approaching a surface **210a** of a plate **210**. For example, the first distance d1 between the surface **220a** of the storing member **220** and a first center c1 of the first magnet **225a** may be different from the second distance d2 between the surface **220a** of the storing member **220** and a second center c2 of the second magnet **131** in the electronic pen **130**. For example, the first distance d1 between the surface **220a** of the storing member **220** and the first center c1 of the first magnet **225a** may be greater than or equal to the second distance d2 between the surface **220a** of the storing member **220** and the second center c2 of the second magnet **131** in the electronic pen **130**. The first center c1 may be positioned on a center point of a first side surface **225a-1** of the first magnet **225a** facing the fourth direction (e.g., the +z direction) and a second side surface **225a-2** of the first magnet **225a** opposite to the first side surface **225a-1**. The second side surface **225a-2**

of the first magnet **225a** may face the third direction (e.g., the $-z$ direction). The second center **c2** may be positioned on a center point of a first side surface **131a** of the second magnet **131** facing the fourth direction (e.g., the $+z$ direction) and a second side surface **131b** of the second magnet **131** opposite to the first side surface **131a**. The second side surface **131b** of the second magnet **131** may face the third direction (e.g., the $-z$ direction). However, it is not limited thereto. For example, the first distance **d1** may mean a distance between the surface **220a** of the storing member **220** and the side surface **225a-1** of the first magnet **225a** facing the fourth direction (e.g., the $+z$ direction), and the second distance **d2** may mean a distance between the surface **220a** of the storing member **220** and the side surface **131a** of the second magnet **131** in the electronic pen **130** facing the fourth direction (e.g., the $+z$ direction). For example, the first distance **d1** between the surface **220a** of the storing member **220** and the side surface **225a-1** of the first magnet **225a** facing the fourth direction (e.g., the $+z$ direction) may be greater than or equal to the second distance **d2** between the surface **220a** of the storing member **220** and the side surface **131a** of the second magnet **131** in the electronic pen **130** facing the fourth direction. For example, as the first magnet **225a** is supported by a supporting member **227** disposed in the second protruding portion **223**, the first distance **d1** may be greater than or equal to the second distance **d2**. For example, in case that the second distance **d2** is smaller than the first distance **d1**, the electronic pen **130** may receive a force in a direction inclined counterclockwise with respect to the first direction (e.g., the $+y$ direction) by the first actuating portion **225** while being seated on the storing member **220**. In case that the electronic pen **130** receives the force in the direction inclined counterclockwise with respect to the first direction (e.g., the $+y$ direction), the end **130a** of the electronic pen **130** may not move into the receiving space **224**. Since the first distance **d1** has a size greater than or equal to the second distance **d2**, the case **200** according to yet another embodiment may provide a structure in which the end **130a** of the electronic pen **130** is easily inserted into the receiving space **224**. For example, in case that the first distance **d1** is greater than the second distance **d2**, the first actuating portion **225** may provide the electronic pen **130** with a force inclined clockwise with respect to the first direction (e.g., the $+y$ direction) while the electronic pen **130** is seated on the storing member **220**. As the electronic pen **130** is provided with the force inclined clockwise with respect to the first direction (e.g., $+y$ direction), the end **130a** of the electronic pen **130** may be easily inserted into the receiving space **224**.

[0052] According to yet another embodiment, a second actuating portion **226** may include at least one third magnet **226a**. The at least one third magnet **226a** may be disposed in a seating portion **221**. The at least one third magnet **226a** may be disposed on the surface **210a** of the plate **210**. The at least one third magnet **226a** may be disposed in the seating portion **221**. The at least one third magnet **226a** may be disposed between the first actuating portion **225** and the receiving space **224**. The at least one third magnet **226a** may be disposed between the end **130a** of the electronic pen **130** and the other end **130b** of the electronic pen **130** in a state in which the electronic pen **130** is seated on the storing member **220**. The at least one third magnet **226a** may include a plurality of third magnets spaced apart from each other.

[0053] According to yet another embodiment, the electronic pen **130** may include at least one fourth magnet **132** corresponding to the at least one third magnet **226a**. The at least one fourth magnet **132** may be disposed in the electronic pen **130**. For example, the at least one fourth magnet **132** may be disposed between the end **130a** of the electronic pen **130** and the other end **130b** of the electronic pen **130**. According to yet another embodiment, the number of the at least one fourth magnet **132** may be substantially the same as the number of the at least one third magnet **226a**. For example, the at least one fourth magnet **132** may include a plurality of fourth magnets spaced apart from each other. However, it is not limited thereto. For example, the number of the at least one fourth magnet **132** may be different from the number of the at least one third magnet **226a**.

[0054] According to yet another embodiment, the second actuating portion **226** may detachably attach the electronic pen **130** on the surface **220a** of the storing member **220**. For example, the at

least one third magnet **226a** of the second actuating portion **226** may attach the electronic pen **130** on the surface **220a** of the storing member **220** by interacting with the at least one fourth magnet **132** in the electronic pen **130**. For example, the at least one third magnet **226a** may apply a force in the third direction (e.g., the $-z$ direction) to the electronic pen **130** by exerting an attractive force to the at least one fourth magnet **132** disposed in the electronic pen **130**. As the force in the third direction (e.g., the $-z$ direction) is applied to the electronic pen **130**, the electronic pen **130** may maintain a state of being attached on the surface **220a** of the storing member **220**.

[0055] According to yet another embodiment, magnitude of a force applied between the first magnet **225a** and the second magnet **131** may be greater than magnitude of a force applied between the at least one third magnet **226a** and the at least one fourth magnet **132**. As the magnitude of the force applied between the first magnet **225a** and the second magnet **131** is relatively greater, when the electronic pen **130** is inserted into the receiving space **224**, the electronic pen **130** may move along the first direction (e.g., the $+y$ direction) while the at least one third magnet **226a** and the at least one fourth magnet **132** interact.

[0056] According to yet another embodiment, the seating portion **221**, which is a portion of the storing member **220**, may have a shape that is open toward the outside of the storing member **220**. For example, the seating portion **221** may be opened toward the outside of the storing member **220** by having a thickness thinner than a first protruding portion **222** and the second protruding portion **223**. The seating portion **221** may have a first thickness $t1$. The first protruding portion **222** may have a second thickness $t2$ greater than the first thickness $t1$. The second protruding portion **223** may have a third thickness $t3$ greater than the first thickness $t1$. As the seating portion **221** has the open shape, at least a portion of the electronic pen **130** may be exposed to the outside of the storing member **220** in a state of being seated in the storing member **220**. For example, at least a portion of the electronic pen **130** may not be covered by the storing member **220** in the state of being seated in the storing member **220**. According to yet another embodiment, the second thickness $t2$ and the third thickness $t3$ may be substantially the same. For example, the second thickness $t2$ may be substantially the same as a sum of the first thickness $t1$ and a fourth thickness $t4$, which is a thickness of the electronic pen **130**. For example, the third thickness $t3$ may be substantially the same as the sum of the first thickness $t1$ and the fourth thickness $t4$, which is the thickness of the electronic pen **130**. However, it is not limited thereto. For example, the second thickness $t2$ and the third thickness $t3$ may be different from each other.

[0057] As described above, the case **200** according to yet another embodiment may provide a structure in which the electronic pen **130** may be easily inserted into the receiving space **224** by the first actuating portion **225** that transmits the force in the first direction (e.g., the $+y$ direction) to the other end **130b** of the electronic pen **130**.

[0058] FIG. **3A** is a perspective view of a case according to an embodiment of the disclosure.

[0059] FIG. **3B** is a cross-sectional view illustrating an example of a case cut along line B-B' of FIG. **1B** according to an embodiment of the disclosure.

[0060] Referring to FIGS. **3A** and **3B**, in a first state of a case **200**, a first portion **211** may form a step with respect to a third portion **213**. In the first state of the case **200**, a step may be formed at a boundary between the first portion **211** and the third portion **213**. In the first state of the case **200**, a second portion **212** may form a step with respect to the third portion **213**. In the first state of the case **200**, a step may be formed at a boundary between the second portion **212** and the third portion **213**.

[0061] According to an embodiment, a cross-sectional area of a receiving space **224** may be smaller than a cross-sectional area of an electronic pen **130**. For example, a width of the receiving space **224** may be smaller than a width of the electronic pen **130**. A width of a component may mean a distance in a direction (e.g., a $+x$ direction or a $-x$ direction) perpendicular to a first direction (e.g., a $+y$ direction) and a third direction (e.g., a $-z$ direction), and the corresponding expression may be utilized substantially the same below unless otherwise stated. As the cross-sectional area of the

receiving space **224** is smaller than the cross-sectional area of the electronic pen **130**, a movement range of the electronic pen **130** inserted into the receiving space **224** by a first actuating portion (e.g., the first actuating portion **225** of FIG. **1C**) may be reduced (or limited). As the movement range of the electronic pen **130** inserted into the receiving space **224** is reduced (or limited), at least a portion of the electronic pen **130** may be exposed to the outside of a storing member **220** such that a user may easily grip the electronic pen **130**.

[0062] According to another embodiment, the receiving space **224** may include a first region **224a** and a second region **224b**. The first region **224a** may receive a portion of the electronic pen **130**. The portion of the electronic pen **130** may be inserted into the first region **224a**. The first region **224a** may have a shape corresponding to a shape of the electronic pen **130**. For example, in case that a cross section of the electronic pen **130** has a substantially circular shape, a cross section of the first region **224a** may have a substantially circular shape. The second region **224b** may be connected to the first region **224a**. A size of the second region **224b** may be smaller than a size of the first region **224a**. For example, a width of the second region **224b** may be smaller than a width of the first region **224a**. As the size of the second region **224b** is smaller than the size of the first region **224a**, movement of the electronic pen **130** in the third direction (e.g., the $-z$ direction) and/or a fourth direction (e.g., a $+z$ direction) may be limited.

[0063] According to yet another embodiment, a surface **210a** of a plate **210** may be closer to the second region **224b** from among the first region **224a** and the second region **224b**. For example, a distance from the surface **210a** of the plate **210** to the first region **224a** may be greater than a distance from the surface **210a** of the plate **210** to the second region **224b**.

[0064] Referring to FIG. **3B**, according to an embodiment, a seating portion **221** may include a protrusion **221a**. The protrusion **221a** may reduce a movement range of the electronic pen **130** with respect to the storing member **220**. For example, the protrusion **221a** may reduce movement of the electronic pen **130** to the outside of the storing member **220**. The protrusion **221a** may protrude from a surface **220a** of the storing member **220** on which the electronic pen **130** is disposed. For example, the protrusion **221a** may extend along the fourth direction (e.g., the $+z$ direction) from the surface **220a** of the storing member **220**. At least a portion of the protrusion **221a** may be in contact with at least a portion of the electronic pen **130** disposed on the surface **220a** of the storing member **220**. For example, at least a portion of the protrusion **221a** may be in contact with a periphery of at least a portion of the electronic pen **130**.

[0065] Referring back to FIG. **3A**, according to an embodiment, a first protruding portion **222** may include a friction member **222a**. The friction member **222a** may limit the movement of the electronic pen **130** to the outside of the receiving space **224**. The friction member **222a** may be in contact with a portion of the electronic pen **130** inserted into the receiving space **224**. For example, the friction member **222a** may be in contact with an outer surface of the portion of the electronic pen **130** inserted into the receiving space **224**. As the electronic pen **130** is provided with a friction force due to contact with the friction member **222a**, movement of the electronic pen **130** may be limited. For example, the friction member **222a** may be manufactured from a material (e.g., thermoplastic polyurethane (TPU), and/or silicone) for providing a friction force to the electronic pen **130**. The friction member **222a** may be disposed inside the receiving space **224**. For example, the friction member **222a** may be disposed along an inner surface of the receiving space **224**. For example, the friction member **222a** may be disposed on an inner surface of the first protruding portion **222** along a periphery of the receiving space **224**.

[0066] As described above, since the size of the second region **224b** is smaller than the size of the first region **224a** of the receiving space **224**, the case **200** according to another embodiment may provide a structure in which a position of the electronic pen **130** inserted into the receiving space **224** may be maintained. The case **200** according to yet another embodiment may provide a structure in which a position of the electronic pen **130** seated on the storing member **220** may be maintained by the protrusion **221a** and the friction member **222a** that reduce the movement range

of the electronic pen **130**.

[0067] FIG. **4A** is a top plan view of a case according to an embodiment of the disclosure.

[0068] FIG. **4B** is a cross-sectional view illustrating an example of a case cut along line C-C' of FIG. **4A** according to an embodiment of the disclosure.

[0069] Referring to FIGS. **4A** and **4B**, according to an embodiment, a storing member **220** may include a barrier **228**. The barrier **228** may limit movement of an electronic pen **130** with respect to the storing member **220**. The barrier **228** may reduce a movement range of the electronic pen **130** with respect to the storing member **220**. According to another embodiment, the barrier **228** may be disposed on a surface **210a** of a plate **210**. The barrier **228** may be connected to a seating portion **221**. The barrier **228** may be connected to at least one of a first protruding portion **222** and a second protruding portion **223**. For example, the barrier **228** may be connected to the first protruding portion **222**. For example, the barrier **228** may be connected to the second protruding portion **223**. For example, the barrier **228** may be connected to the first protruding portion **222** and the second protruding portion **223**. A length of the barrier **228** may be less than or equal to a length of the electronic pen **130**. For example, the length of the barrier **228** may be shorter than a length of the seating portion **221**. For example, the length of the barrier **228** may be substantially the same as the length of the seating portion **221**.

[0070] According to yet another embodiment, the barrier **228** may shield a magnetic field from a third magnet **226a** of a second actuating portion **226** and/or a fourth magnet **132** in the electronic pen **130**. For example, the barrier **228** may include a conductive material (e.g., a metallic material). As the barrier **228** shields the magnetic field from at least one of the third magnet **226a** and the fourth magnet **132**, malfunction of an electronic device **100** disposed on a case **200** may be reduced.

[0071] According to yet another embodiment, a thickness of the barrier **228** may be thicker than a thickness of the seating portion **221**. For example, the thickness of the barrier **228** may be substantially the same as a sum of the thickness of the seating portion **221** and a thickness of the electronic pen **130**. For example, the thickness of the barrier **228** may be smaller than the sum of the thickness of the seating portion **221** and the thickness of the electronic pen **130**. As the barrier **228** is disposed, movement of the electronic pen **130** to the outside of the storing member **220** may be reduced (or limited).

[0072] Meanwhile, in FIGS. **4A** and **4B**, the barrier **228** is illustrated as being connected to a side surface of the seating portion **221** facing a +x-axis direction, but is not limited thereto. For example, the barrier **228** may be connected to another side surface of the seating portion **221** facing a -x-axis direction. For example, the barrier **228** may be connected to the side surface of the seating portion **221** facing the +x-axis direction and the other side surface of the seating portion **221** facing the -x-axis direction.

[0073] As described above, the case **200** according to yet another embodiment may provide a structure in which a position of the electronic pen **130** may be maintained by the barrier **228** that limits the movement range of the electronic pen **130** with respect to the storing member **220**.

[0074] FIG. **5** is a cross-sectional view illustrating an example of a case according to an embodiment of the disclosure.

[0075] Referring to FIG. **5**, according to an embodiment, a first actuating portion **225** may include a pressing member **225b** (or a pressing portion **225b**) and an elastic member **225c**. The first pressing portion **225b** may be movable along a first direction (e.g., a +y direction) or a second direction (e.g., a -y direction) opposite to the first direction with respect to a storing member **220**. For example, the first pressing portion **225b** may move along the first direction (e.g., the +y direction) such that an end **130a** of an electronic pen **130** is inserted into a receiving space **224**. According to an embodiment, at least a portion of the first pressing portion **225b** may be disposed in the storing member **220**. The first pressing portion **225b** may protrude from the inside of the storing member **220** to the outside of the storing member **220** by moving in the first direction (e.g.,

the +y direction). The first pressing portion **225b** may be in contact with another end **130b** of the electronic pen **130** by protruding to the outside of the storing member **220**. The first pressing portion **225b** may apply a force in the first direction (e.g., the +y direction) to the electronic pen **130** by being in contact with the other end **130b** of the electronic pen **130**.

[0076] According to another embodiment, the elastic member **225c** may provide a force to the pressing member **225b**. The elastic member **225c** may depress the pressing member **225b**. For example, the elastic member **225c** may provide a force in the first direction (e.g., the +y direction) to the pressing member **225b** by being in contact with the pressing member **225b**. For example, the elastic member **225c** may provide an elastic force to the pressing member **225b** by elastically deforming. For example, the elastic member **225c** may be a spring, but is not limited thereto. For example, the elastic member **225c** may be replaced with various members that follow Hooke's Law. According to yet another embodiment, the elastic member **225c** may be disposed in the storing member **220**. The elastic member **225c** may be disposed in a second protruding portion **223** of the storing member **220**.

[0077] According to yet another embodiment, a supporting member **227** may support the pressing member **225b** of the first actuating portion **225**. For example, the supporting member **227** may be disposed between the pressing member **225b** and a surface **210a** of a plate **210**. The supporting member **227** may support the elastic member **225c** of the first actuating portion **225**. For example, the supporting member **227** may be disposed between the elastic member **225c** and the surface **210a** of the plate **210**.

[0078] According to yet another embodiment, the electronic pen **130** may include a receiving groove **133**. The receiving groove **133** may be disposed in the other end **130b** of the electronic pen **130**. For example, as a portion of the other end **130b** of the electronic pen **130** is recessed inwardly into the electronic pen **130**, the receiving groove **133** may be formed. The receiving groove **133** may receive a portion of the pressing member **225b**. For example, the receiving groove **133** may have a shape corresponding to a portion of the pressing member **225b**. For example, while the pressing member **225b** protrudes outside the second protruding portion **223**, the electronic pen **130** may approach the storing member **220**. As the electronic pen **130** approaches the storing member **220**, the other end **130b** of the electronic pen **130** may contact the pressing member **225b**. The pressing member **225b** may move inside the storing member **220** along the second direction (e.g., the -y direction) by contact with the other end **130b** of the electronic pen **130**. As the pressing member **225b** moves in the second direction (e.g., the -y direction), the elastic member **225c** may be compressed. While the pressing member **225b** is inserted into the receiving groove **133** of the electronic pen **130**, the compressed elastic member **225c** may depress the pressing member **225b** in the first direction (e.g., the +y direction). The pressing member **225b** may move the electronic pen **130** in the first direction (e.g., the +y direction) based on receiving a force from the elastic member **225c**. The electronic pen **130** may be inserted into the receiving space **224** by moving along the first direction (e.g., the +y direction) by the first pressing portion **225b**. According to yet another embodiment, the pressing member **225b** may apply a force in the first direction (e.g., the +y direction) to the other end **130b** of the electronic pen **130** while the electronic pen **130** is inserted into the receiving space **224**. As the pressing member **225b** provides the force in the first direction (e.g., the +y direction) to the electronic pen **130**, a position of the electronic pen **130** may maintain a state of being inserted into the receiving space **224**.

[0079] According to yet another embodiment, a third distance $d3$ between a surface **220a** of the storing member **220** on which the electronic pen **130** is disposed and the pressing member **225b** may be different from half of a fourth thickness $t4$, which is a thickness of the electronic pen **130**. For example, the third distance $d3$ may be greater than or equal to half of the fourth thickness $t4$. In case that the third distance $d3$ is less than half of the fourth thickness $t4$, the electronic pen **130** may receive a force inclined counterclockwise with respect to the first direction (e.g., the +y direction) while being seated on the storing member **220**. In case that the electronic pen **130** receives the force

inclined counterclockwise with respect to the first direction (e.g., the +y direction), the end **130a** of the electronic pen **130** may not move into the receiving space **224**. Since the third distance **d3** is greater than or equal to half of the fourth thickness **14**, a case **200** according to yet another embodiment may provide a structure in which the end **130a** of the electronic pen **130** is easily inserted into the receiving space **224**.

[0080] As described above, the case **200** according to yet another embodiment may provide a structure in which the electronic pen **130** is easily seated on the storing member **220** by the first actuating portion **225** including the pressing member **225b** and the elastic member **225c**.

[0081] A case may include a structure for receiving an electronic pen for transmitting a user input to an electronic device. When the user uses the electronic pen, the electronic pen may be separated from the case. The user may detachably couple the electronic pen to the case after using the electronic pen. When the user couples the electronic pen to the case, a structure for easily receiving the electronic pen in the case may be required. When the user carries the case in a state of receiving the electronic pen in the case, a structure in which the electronic pen is not easily separated from the case may be required.

[0082] According to yet another embodiment, a case (e.g., the case **200** of FIGS. **1A**, **1B**, and **1C**) detachably coupled to an electronic device (e.g., the electronic device **100** of FIGS. **1A** and **1B**) is provided. According to yet another embodiment, the case may comprise a plate (e.g., the plate **210** of FIGS. **1A**, **1B**, and **1C**) supporting the electronic device. According to yet another embodiment, the case may comprise a storing member (e.g., the storing member **220** of FIGS. **1A**, **1B**, and **1C**) for storing at least a portion of an electronic pen (e.g., the electronic pen **130** of FIGS. **1A**, **1B**, and **1C**) disposed on a surface (e.g., the surface **210a** of FIGS. **1A**, **1B**, and **1C**) of the plate. According to yet another embodiment, the storing member may include a receiving space (e.g., the receiving space **224** of FIGS. **1A** and **1C**) receiving an end (e.g., the end **130a** of FIGS. **1A** and **1C**) of the electronic pen. According to yet another embodiment, the storing member may include a first actuating portion (e.g., the first actuating portion **225** of FIG. **1C**) applying a force in a direction toward the receiving space to another end (e.g., the other end **130b** of FIGS. **1A** and **1C**) of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space. According to yet another embodiment, the storing member may include a second actuating portion (e.g., the second actuating portion **226** of FIG. **1C**), disposed between the receiving space and the first actuating portion, and applying a force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member. According to yet another embodiment, a portion of the storing member may open toward an outside of the storing member such that at least a portion of the electronic pen is exposed outside the storing member.

[0083] The case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space by the first actuating portion that transmits a force in a direction toward the receiving space to the other end of the electronic pen. The electronic pen according to yet another embodiment may provide a structure in which the electronic pen is easily coupled to the case by the second actuating portion that provides a force for seating the electronic pen on the storing member. Since the storing member is open toward the outside, the electronic device according to yet another embodiment may provide a structure in which a user may easily separate the electronic pen coupled to the case.

[0084] According to yet another embodiment, the first actuating portion may include a first magnet (e.g., the first magnet **225a** of FIG. **2**). According to yet another embodiment, the first magnet may exert a repulsive force to a second magnet (e.g., the second magnet **131** of FIG. **2**) disposed inside the another end of the electronic pen, thereby inserting the end of the electronic pen into the receiving space.

[0085] The case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space by the first magnet that transmits a

force in a direction toward the receiving space to the other end of the electronic pen.

[0086] According to yet another embodiment, a distance (e.g., the first distance d1 of FIG. 2) between a surface of the storing member on which the electronic pen is disposed and a center (e.g., the first center c1 of FIG. 2) of the first magnet may be greater than or equal to a distance (e.g., the second distance d2 of FIG. 2) between the surface of the storing member and a center (e.g., the second center c2 of FIG. 2) of the second magnet.

[0087] According to yet another embodiment, a distance (e.g., the first distance d1 of FIG. 2) between a side surface (e.g., the side surface 225a-1 of FIG. 2) of the first magnet facing a direction in which a surface (e.g., the surface 220a of FIG. 2) of the storing member on which the electronic pen is disposed faces and the surface of the storing member may be greater than or equal to a distance (e.g., the second distance d2 of FIG. 2) between a side surface (e.g., the side surface 131a of FIG. 2) of the second magnet facing a direction in which the surface of the storing member faces and the surface of the storing member.

[0088] Since the first distance between the storing member and the first magnet is greater than or equal to a distance between the storing member and the second magnet, the case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space.

[0089] According to yet another embodiment, the storing member may include a supporting member (e.g., the supporting member 227 of FIG. 2) supporting the first magnet by being disposed between the first magnet and the surface of the plate.

[0090] Since the first distance between the storing member and the first magnet is greater than or equal to the distance between the storing member and the second magnet, by the supporting member supporting the first magnet, the case according to yet another embodiment may provide the structure in which the electronic pen may be easily inserted into the receiving space.

[0091] According to yet another embodiment, the first actuating portion may include a pressing member (e.g., the pressing member 225b of FIG. 5) that moves in a first direction such that the another end of the electronic pen is inserted into the receiving space or in a second direction opposite to the first direction. According to yet another embodiment, the first actuating portion may include an elastic member (e.g., the pressing member 225b of FIG. 5), disposed in the storing member, depressing the pressing member.

[0092] The case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space by the pressing member that transmits a force in a direction toward the receiving space to the other end of the electronic pen.

[0093] According to yet another embodiment, a distance (e.g., the third distance d3 of FIG. 5) between a surface of the storing member on which the electronic pen is disposed and the pressing member may be greater than or equal to half a thickness (e.g., the fourth thickness t4 of FIG. 5) of the electronic pen.

[0094] Since the third distance between the storing member and the pressing member is greater than or equal to half of the thickness of the electronic pen, the case according to yet another embodiment may provide the structure in which the electronic pen may be easily inserted into the receiving space.

[0095] According to yet another embodiment, the storing member may include a supporting member (e.g., the supporting member 227 of FIG. 5) disposed between the surface of the plate and the pressing member.

[0096] Since the third distance between the storing member and the pressing member is greater than or equal to half of the thickness of the electronic pen, by the supporting member supporting the first magnet, the case according to yet another embodiment may provide the structure in which the electronic pen may be easily inserted into the receiving space.

[0097] According to yet another embodiment, the storing member may include a seating portion (e.g., the seating portion 221 of FIG. 1C) on which the electronic pen is seated. According to an

embodiment, the storing member may include a first protruding portion (e.g., the first protruding portion **222** of FIG. **1C**), connected to an end of the seating portion, surrounding the receiving space, and having a greater thickness than the seating portion. According to yet another embodiment, the storing member may include a second protruding portion (e.g., the second protruding portion **223** of FIG. **1C**), connected to another end of the seating portion opposite to the end of the seating portion, surrounding the first actuating portion, and having a thickness greater than a thickness of the seating portion.

[0098] The case according to yet another embodiment may provide a structure in which a portion of the storing member is opened toward the outside by the first protruding portion and the second protruding portion defining a step with respect to the seating portion.

[0099] According to yet another embodiment, the seating portion may include a protrusion (e.g., the protrusion **221a** of FIG. **3B**), protruding from a surface of the storing member on which the electronic pen is disposed, and surrounding the electronic pen.

[0100] The case according to yet another embodiment may provide a structure capable of reducing a movement range of the electronic pen with respect to the storing member by the protrusion formed on the seating portion.

[0101] According to yet another embodiment, the storing member may include a barrier (e.g., the barrier **228** of FIGS. **4A** and **4B**), having a thickness greater than the thickness of the seating portion, and connected to the second protruding portion.

[0102] The case according to yet another embodiment may provide the structure capable of reducing the movement range of the electronic pen with respect to the storing member by the barrier connected to the first protruding portion.

[0103] According to yet another embodiment, the receiving space may include a first region (e.g., the first region **224a** of FIG. **3A**) having a shape corresponding to a shape of the electronic pen. According to yet another embodiment, the receiving space may include a second region (e.g., the second region **224b** of FIG. **3A**), having a width smaller than a width of the first region, connected to the first region. According to yet another embodiment, the surface of the plate may be closer to the second region from among the first region and the second region.

[0104] Since the second region close to the plate is smaller than the first region far from the plate, the case according to yet another embodiment may provide the structure capable of reducing the movement range of the electronic pen.

[0105] According to yet another embodiment, the plate may include a first portion (e.g., the first portion **211** of FIG. **1C**) on which the electronic device is disposed. According to yet another embodiment, the plate may include a second portion (e.g., the second portion **212** of FIG. **1C**) being foldable with respect to the first portion. According to yet another embodiment, the plate may include a third portion (e.g., the third portion **213** of FIG. **1C**) disposed between the first portion and the second portion. According to yet another embodiment, the storing member may be disposed on the third portion.

[0106] The case according to yet another embodiment may provide a structure capable of covering the electronic device by the second portion movable with respect to the first portion.

[0107] According to yet another embodiment, a state of the case may include a first state in which a direction in which an inner surface (e.g., the inner surface **211a** of FIG. **1C**) of the first portion facing the electronic device faces and a direction in which an inner surface (e.g., the inner surface **212a** of FIG. **1C**) of the second portion faces are the same, and a second state in which the inner surface of the first portion and the inner surface of the second portion face each other. According to yet another embodiment, in the first state, a step may be formed at a boundary between the first portion and the third portion.

[0108] The case according to yet another embodiment may provide the structure capable of covering the electronic device by the second portion movable with respect to the first portion.

[0109] According to yet another embodiment, the second actuating portion may include at least one

third magnet (e.g., the at least one third magnet **226a** of FIGS. 3A and 3B) disposed between the first actuating portion and the receiving space. According to yet another embodiment, the at least one third magnet may exert an attractive force to at least one fourth magnet (e.g., the at least one fourth magnet **132** of FIGS. 3A and 3B) that corresponds to the at least one third magnet and is disposed in the electronic pen.

[0110] The case according to yet another embodiment may provide a structure capable of maintaining a position of the electronic pen on the storing member by the at least one third magnet disposed between the end of the electronic pen and the other end of the electronic pen.

[0111] According to yet another embodiment, the case may comprise a friction member (e.g., the friction member **222a** of FIG. 3A), disposed along an inner surface of the receiving space, in contact with a portion of the electronic pen inserted into the receiving space.

[0112] The case according to yet another embodiment may provide the structure capable of reducing the movement range of the electronic pen with respect to the receiving space by the friction member disposed in the receiving space.

[0113] According to yet another embodiment, a case (e.g., the case **200** of FIGS. 1A, 1B, and 1C) detachably coupled to an electronic device (e.g., the electronic device **100** of FIGS. 1A and 1B) is provided. According to yet another embodiment, the case may comprise a plate (e.g., the plate **210** of FIGS. 1A, 1B, and 1C) supporting the electronic device. According to yet another embodiment, the case may comprise a storing member (e.g., the storing member **220** of FIGS. 1A, 1B, and 1C) for storing at least a portion of an electronic pen (e.g., the electronic pen **130** of FIGS. 1A, 1B, and 1C) disposed on a surface (e.g., the surface **210a** of FIGS. 1A, 1B, and 1C) of the plate. According to yet another embodiment, the storing member may include a seating portion (e.g., the seating portion **221** of FIG. 1C), having a first thickness (e.g., the first thickness $t1$ of FIG. 2), on which the electronic pen is seated. According to yet another embodiment, the storing member may include a receiving space (e.g., the receiving space **224** of FIG. 1C) receiving an end (e.g., the end **130a** of FIGS. 1A and 1C) of the electronic pen. According to yet another embodiment, the storing member may include a first protruding portion (e.g., the first protruding portion **222** of FIG. 1C), having a second thickness (e.g., the second thickness $t2$ of FIG. 2) greater than the first thickness, surrounding the receiving space, and connected to an end of the seating portion. According to yet another embodiment, the storing member may include a first actuating portion (e.g., the first actuating portion **225** of FIG. 1C) applying a force in a direction toward the receiving space to another end of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space. According to yet another embodiment, the storing member may include a second protruding portion (e.g., the second protruding portion **223** of FIG. 1C), having a third thickness (e.g., the third thickness $t3$ of FIG. 2) greater than the first thickness, surrounding the first actuating portion, and connected to another end (e.g., the other end **130b** of FIGS. 1A and 1C) of the seating portion opposite to the end of the seating portion. According to yet another embodiment, the storing member may include a second actuating portion (e.g., the second actuating portion **226** of FIG. 1C), disposed in the seating portion, applying a force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member.

[0114] The case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space by the first actuating portion that transmits a force in a direction toward the receiving space to the other end of the electronic pen. The electronic pen according to yet another embodiment may provide a structure in which the electronic pen is easily coupled to the case by the second actuating portion that provides a force for seating the electronic pen on the storing member. Since the storing member is open toward the outside, the electronic device according to yet another embodiment may provide a structure in which a user may easily separate the electronic pen coupled to the case.

[0115] According to yet another embodiment, the first actuating portion may include a first magnet

(e.g., the first magnet **225a** of FIG. 2). According to yet another embodiment, the first magnet may exert a repulsive force to a second magnet (e.g., the second magnet **131** of FIG. 2) disposed inside the another end of the electronic pen, thereby inserting the end of the electronic pen into the receiving space.

[0116] The case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space by the first magnet that transmits a force in a direction toward the receiving space to the other end of the electronic pen.

[0117] According to yet another embodiment, the first actuating portion may include a pressing member (e.g., the pressing member **225b** of FIG. 5) that moves in a first direction such that the another end of the electronic pen is inserted into the receiving space or in a second direction opposite to the first direction. According to yet another embodiment, the first actuating portion may include an elastic member (e.g., the pressing member **225b** of FIG. 5), disposed in the storing member, and depressing the pressing member.

[0118] The case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space by the pressing member that transmits a force in a direction toward the receiving space to the other end of the electronic pen.

[0119] According to yet another embodiment, the seating portion may include a protrusion (e.g., the protrusion **221a** of FIG. 3B), protruding from a surface of the storing member on which the electronic pen is disposed, and surrounding the electronic pen.

[0120] The case according to yet another embodiment may provide a structure capable of reducing a movement range of the electronic pen with respect to the storing member by the protrusion formed on the seating portion.

[0121] According to yet another embodiment, the receiving space may include a first region (e.g., the first region **224a** of FIG. 3A) having a shape corresponding to a shape of the electronic pen. According to yet another embodiment, the receiving space may include a second region (e.g., the second region **224b** of FIG. 3A), having a width smaller than a width of the first region, connected to the first region. According to yet another embodiment, the surface of the plate may be closer to the second region from among the first region and the second region.

[0122] Since the second region close to the plate is smaller than the first region far from the plate, the case according to yet another embodiment may provide the structure capable of reducing the movement range of the electronic pen.

[0123] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to yet another embodiment of the disclosure, the electronic devices are not limited to those described above.

[0124] It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” or “connected with” another element (e.g., a second

element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0125] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0126] While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. A case detachably coupled to an electronic device, the case comprising: a plate supporting the electronic device; and a storing member for storing at least a portion of an electronic pen disposed on a surface of the plate, wherein the storing member includes: a receiving space receiving an end of the electronic pen, a first actuating portion applying a force in a direction toward the receiving space to another end of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space, and a second actuating portion, disposed between the receiving space and the first actuating portion, and applying a force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member, and wherein a portion of the storing member is open toward an outside of the storing member such that at least a portion of the electronic pen is exposed outside the storing member.
2. The case of claim 1, wherein the first actuating portion includes a first magnet, and wherein the first magnet exerts a repulsive force to a second magnet disposed inside the another end of the electronic pen, thereby inserting the end of the electronic pen into the receiving space.
3. The case of claim 2, wherein a distance between a surface of the storing member on which the electronic pen is disposed and a center of the first magnet is greater than or equal to a distance between the surface of the storing member and a center of the second magnet.
4. The case of claim 2, wherein the storing member further includes a supporting member supporting the first magnet by being disposed between the first magnet and the surface of the plate.
5. The case of claim 1, wherein the first actuating portion includes: a pressing member that moves in a first direction such that the another end of the electronic pen is inserted into the receiving space or in a second direction opposite to the first direction; and an elastic member, disposed in the storing member, depressing the pressing member.
6. The case of claim 5, wherein a distance between a surface of the storing member on which the electronic pen is disposed and the pressing member is greater than or equal to half a thickness of the electronic pen.
7. The case of claim 5, wherein the storing member further includes a supporting member disposed between the surface of the plate and the pressing member.
8. The case of claim 1, wherein the storing member includes: a seating portion on which the

electronic pen is seated; a first protruding portion, connected to an end of the seating portion, surrounding the receiving space, and having a greater thickness than the seating portion; and a second protruding portion, connected to another end of the seating portion opposite to the end of seating portion, surrounding the first actuating portion, and having a thickness greater than a thickness of the seating portion.

9. The case of claim 8, wherein the seating portion includes a protrusion, protruding from a surface of the storing member on which the electronic pen is disposed, and surrounding the electronic pen.

10. The case of claim 8, wherein the storing member includes a barrier, having a thickness greater than the thickness of the seating portion, and connected to the second protruding portion.

11. The case of claim 1, wherein the receiving space includes: a first region having a shape corresponding to a shape of the electronic pen, and a second region, having a width smaller than a width of the first region, connected to the first region, and wherein the surface of the plate is closer to the second region from among the first region and the second region.

12. The case of claim 1, wherein the plate includes: a first portion on which the electronic device is disposed, a second portion being foldable with respect to the first portion, and a third portion disposed between the first portion and the second portion, and wherein the storing member is disposed on the third portion.

13. The case of claim 12, wherein a state of the case includes: a first state in which a direction in which an inner surface of the first portion facing the electronic device faces and a direction in which an inner surface of the second portion faces are the same, and a second state in which the inner surface of the first portion and the inner surface of the second portion face each other, and wherein, in the first state, a step is formed at a boundary between the first portion and the third portion.

14. The case of claim 1, wherein the second actuating portion includes at least one third magnet disposed between the first actuating portion and the receiving space, and wherein the at least one third magnet exerts an attractive force to at least one fourth magnet that corresponds to the at least one third magnet and is disposed in the electronic pen.

15. The case of claim 1, further comprising: a friction member, disposed along an inner surface of the receiving space, in contact with a portion of the electronic pen inserted into the receiving space.

16. A case detachably coupled to an electronic device, the case comprising: a plate supporting the electronic device; and a storing member for storing at least a portion of an electronic pen disposed on a surface of the plate, wherein the storing member includes: a seating portion, having a first thickness, on which the electronic pen is seated, a receiving space receiving an end of the electronic pen, a first protruding portion, having a second thickness greater than the first thickness, surrounding the receiving space, and connected to an end of the seating portion, a first actuating portion applying a force in a direction toward the receiving space to another end of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space, a second protruding portion, having a third thickness greater than the first thickness, surrounding the first actuating portion, and connected to another end of the seating portion opposite to the end of the seating portion, and a second actuating portion, disposed in the seating portion, applying a force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member.

17. The case of claim 16, wherein the first actuating portion includes a first magnet, and wherein the first magnet exerts a repulsive force to a second magnet disposed inside the another end of the electronic pen, thereby inserting the end of the electronic pen into the receiving space.

18. The case of claim 16, wherein the first actuating portion includes: a pressing member that moves in a first direction such that the another end of the electronic pen is inserted into the receiving space or in a second direction opposite to the first direction; and an elastic member, disposed in the storing member, and depressing the pressing member.

19. The case of claim 16, wherein the seating portion includes a protrusion, protruding from a

surface of the storing member on which the electronic pen is disposed, and surrounding the electronic pen.

20. The case of claim 16, wherein the receiving space includes: a first region having a shape corresponding to a shape of the electronic pen, and a second region, having a width smaller than a width of the first region, connected to the first region, and wherein the surface of the plate is closer to the second region than the first region.
